

① diff b/w Debian & RHEL family.

Debian

RHEL

→ A Linux family based on Debian, focused on stability and free software.	* A Linux family based on Red Hat enterprise Linux, designed for enterprise environments.
Package format: .deb	* .rpm
Package Manager: APT (apt, apt-get)	* DNF (formerly Yum)
popular distros: Debian, Ubuntu, Linux Mint	* RHEL, Rocky Linux
Target users: Beginners, developers, desktops	* Enterprises, business, pro servers.
Support: Community support	* Professional enterprise support of RH
Cost: Mostly free	* RHEL requires a subscription.
Software updates: frequent updates	* Stable, thoroughly tested updates
Commands: apt update, apt install pkg	* dnf update, dnf install package

② about Protocol & Ports in Computer network?

→ A protocol is a set of rules that defines how devices communicate over a Computer network. It specifies how data is sent, received, and interpreted so that different devices can exchange information correctly.

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Common network protocols.

HTTP → transfers web pages.

FTP → transfers file b/w computers.

DNS → converts domain names into IP addresses

SMTP → sends email

POP3 → retrieves emails

→ Port → it is a logical communication endpoint on a computer. it helps the operating system identify which appⁿ or service should receive incoming network data.

ex if you visit a secure website, your computer connects to the server IP address using port 443 (HTTPS).

* Common Port numbers.

<u>Port</u>	<u>Service</u>	<u>Protocol</u>
2021	FTP	TCP
22	SSH	TCP
23	Telnet	TCP
25	SMTP	TCP
53	DNS	TCP/UDP

ex = when you type <https://www.google.com> in your browser.

- (1) DNS (Port 53) finds Google's IP address.
- (2) Your browser connects to Google's server using HTTPS (Port 443)
- (3) The server sends the secure web page back to your browser.